



Centralise or Federate

How to Scale Your Lakehouse



Centralise or Federate

How to Scale Your Lakehouse



Craig Porteous

Technology Director



 <https://craigporteous.com>

 <https://github.com/cporteu>

 <https://www.youtube.com/@cnporteous>





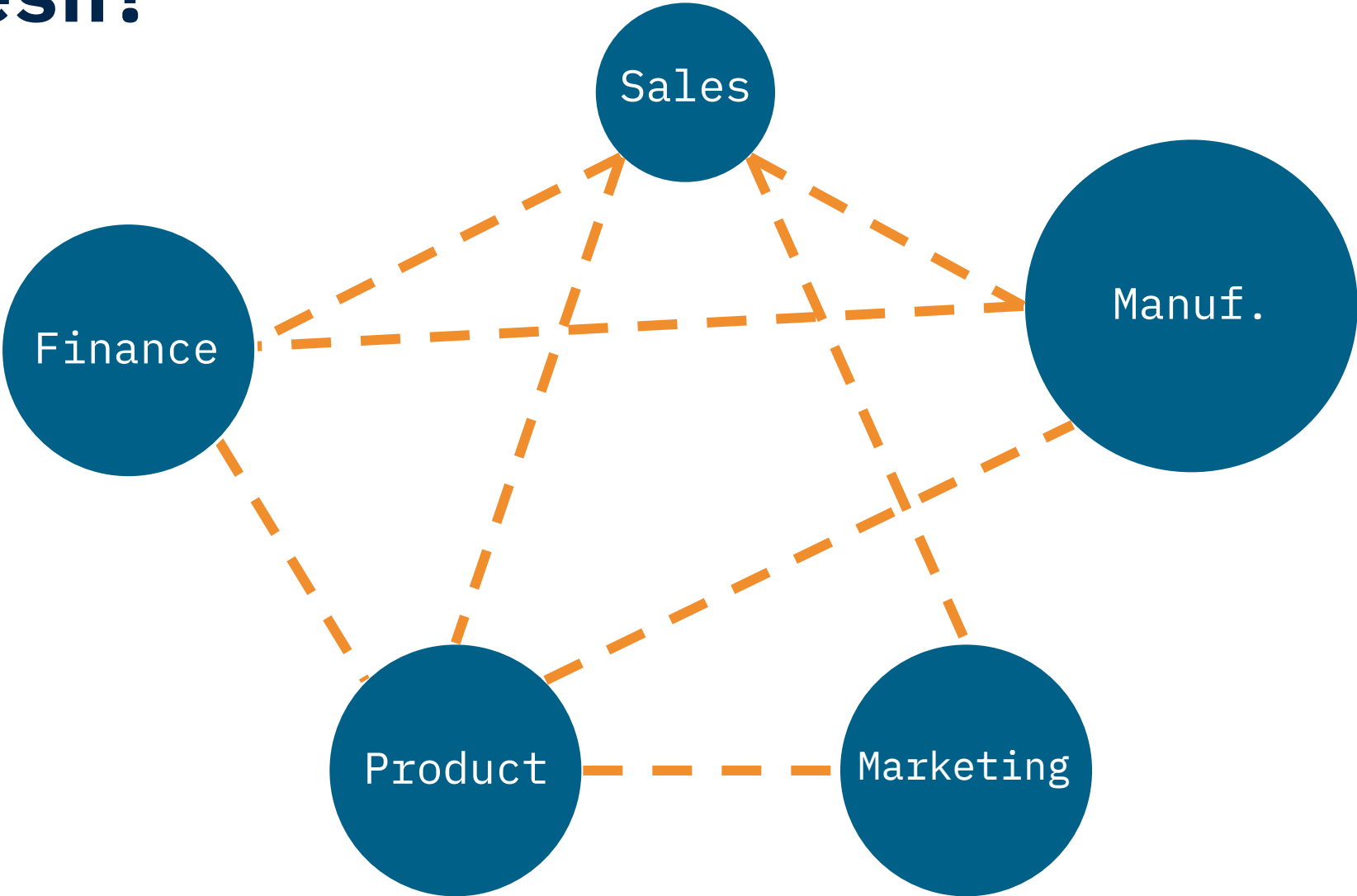




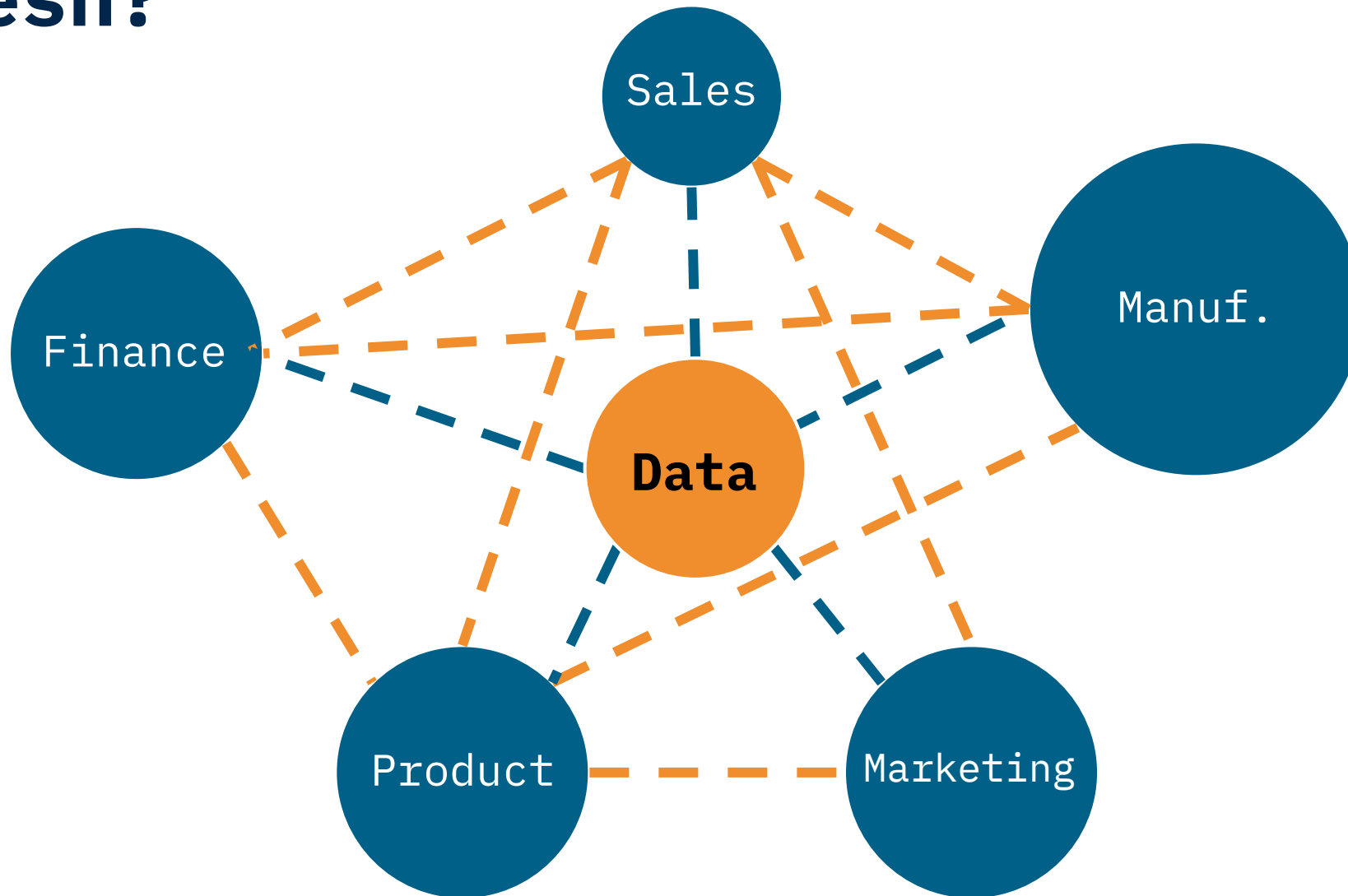
A decorative graphic on the left side of the slide, consisting of a grid of squares in various shades of blue, teal, and orange, arranged in a pattern that tapers off to the right.

What's next?

Data Mesh?



Data Mesh?



Consistency
Control

Data Team



Domain Expertise
Data Quality Ownership

Domains





Consistency
Control

Data Team



Domain Expertise
Data Quality Ownership

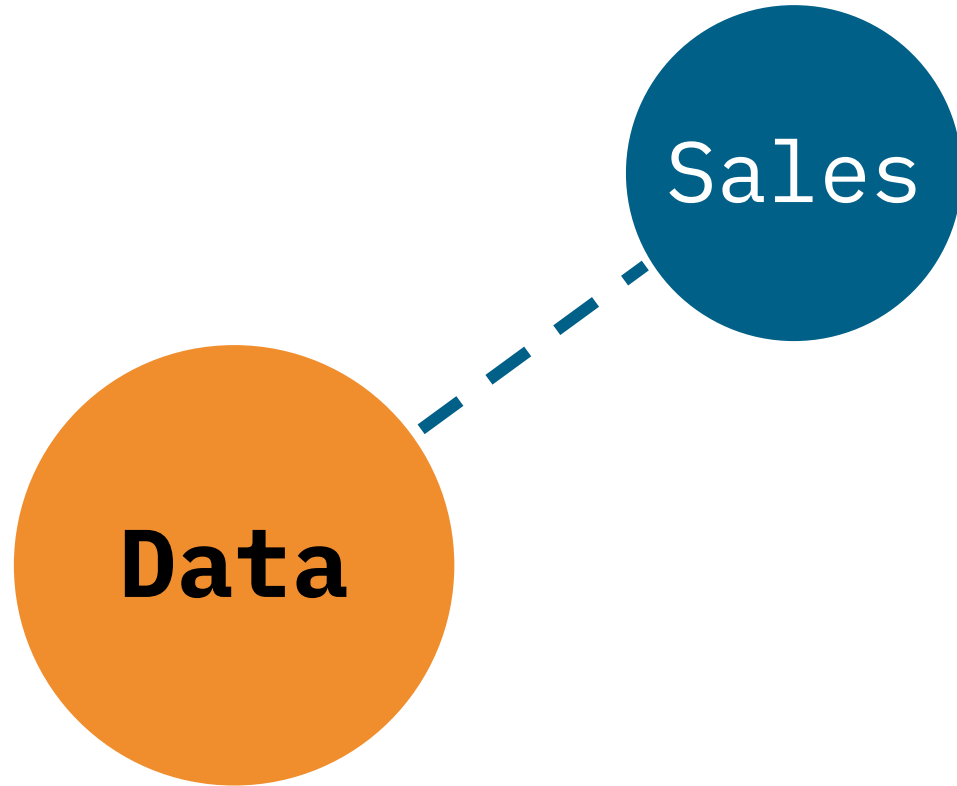
Domains





How?

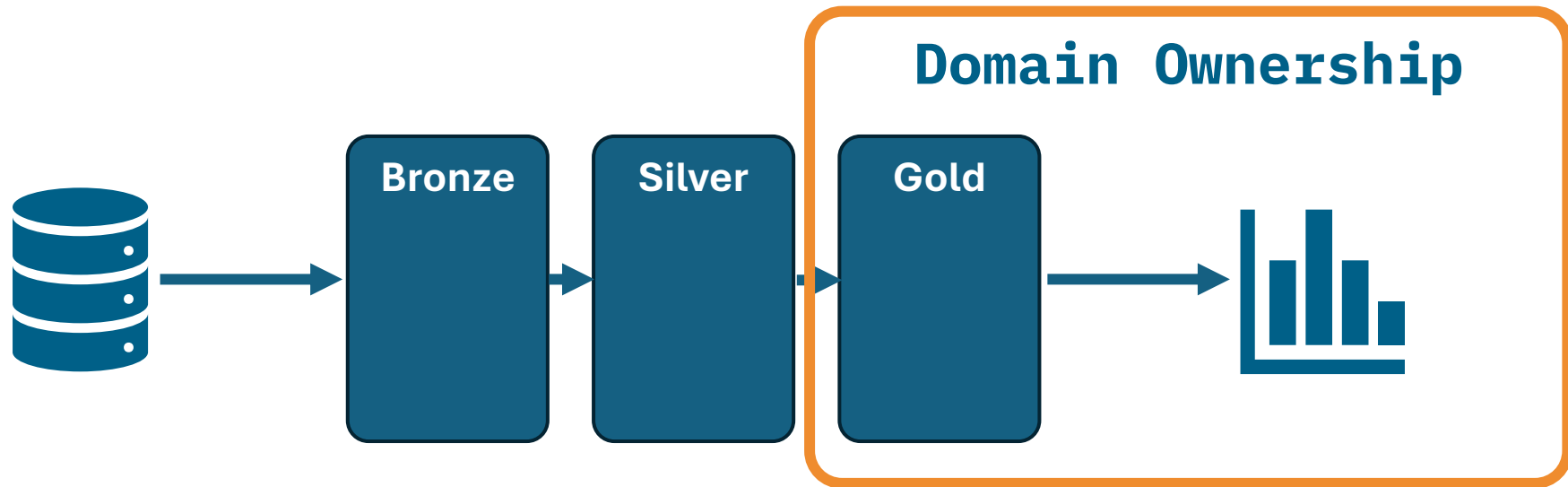
1 Create your Champion



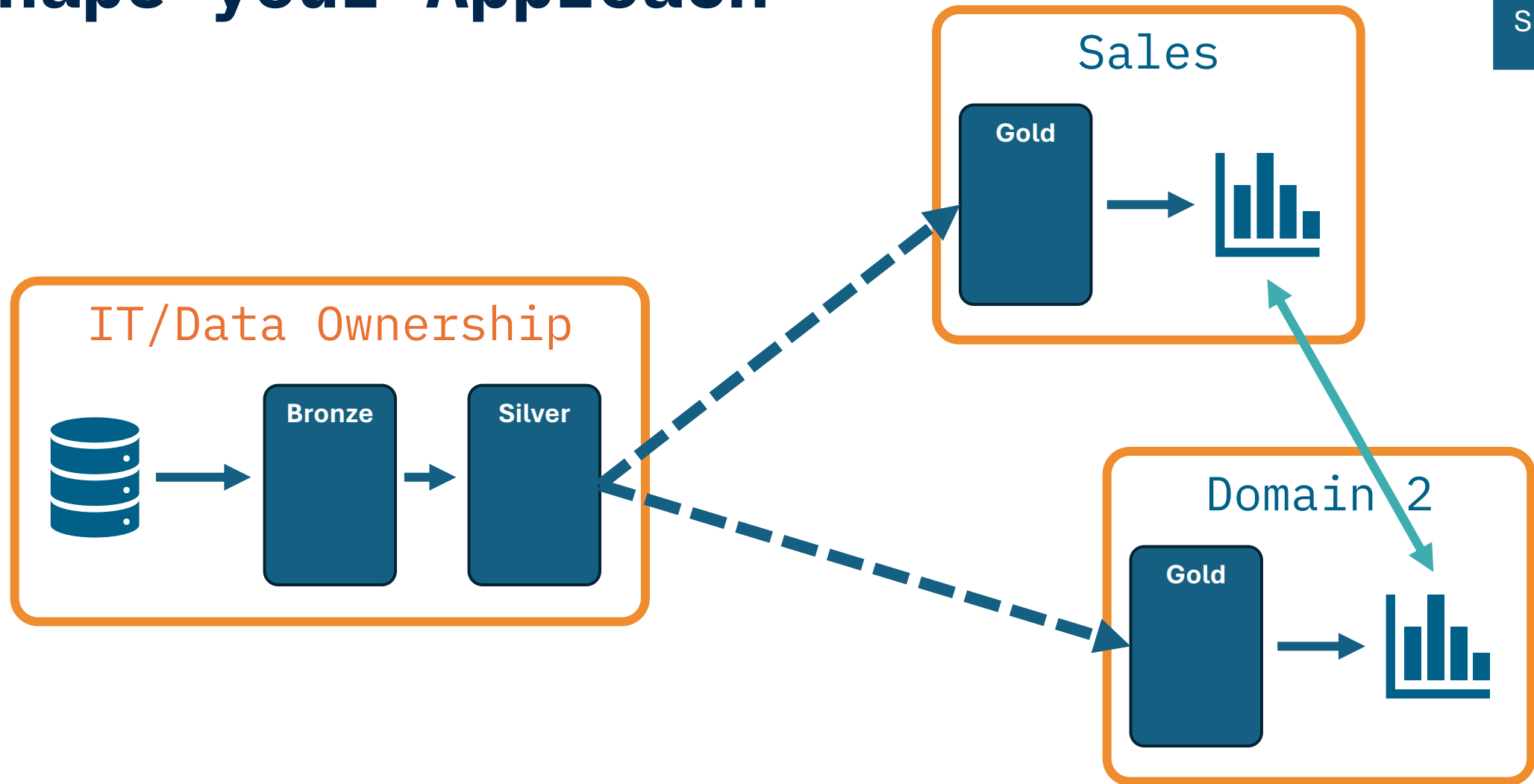
- Understand their needs
- Show them the benefits
- Build a playground
- Support & Train



2 Shape your Approach



2 Shape your Approach



3 Build the Journey

**Assess
Maturity**

Automate

**Provide
Support**



Assess Maturity

Data
Maturity

Maturity

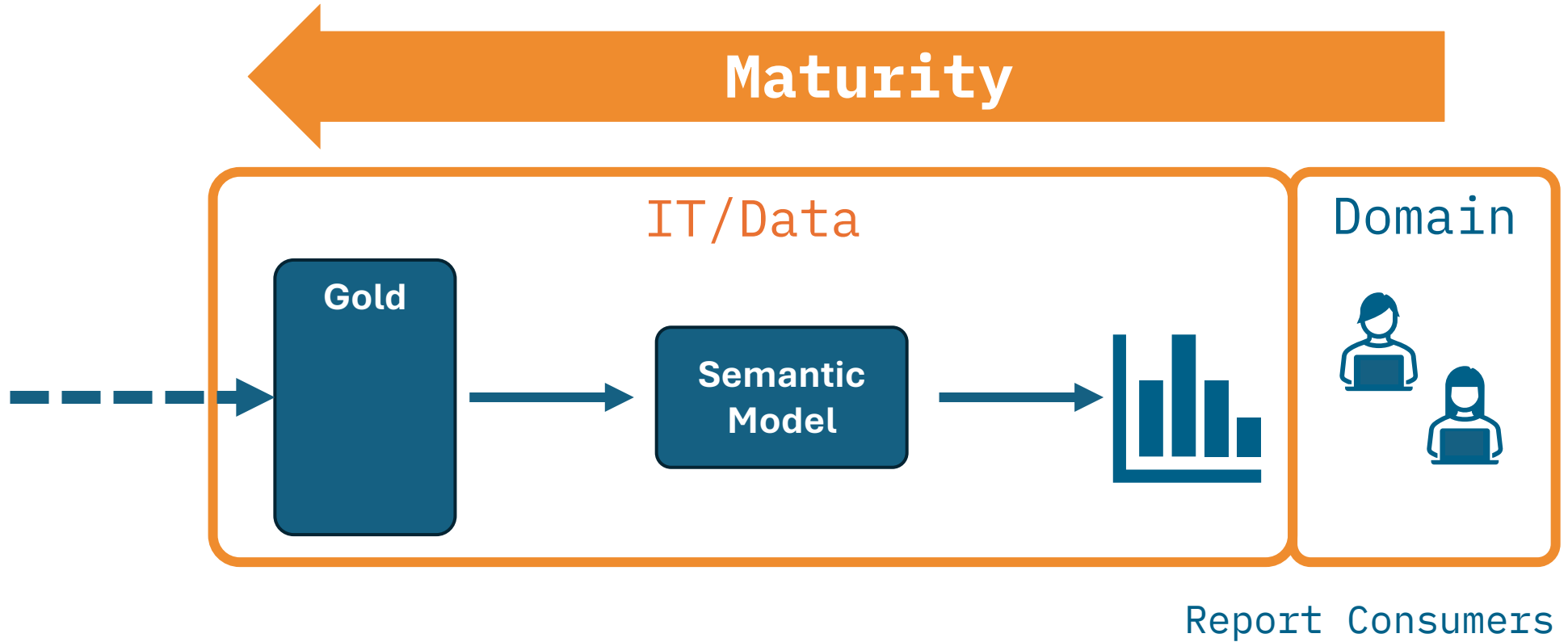
IT/Data

Gold

Semantic
Model

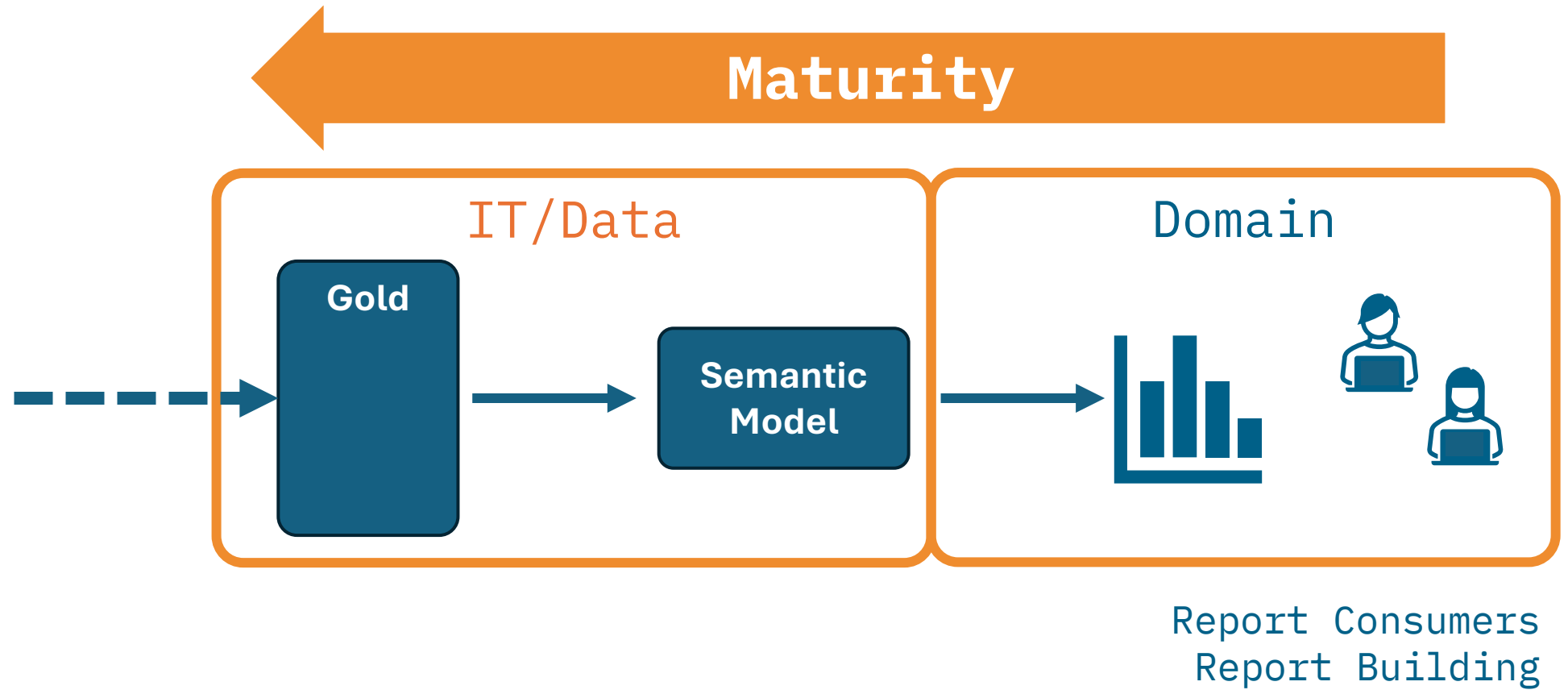
Domain

Report Consumers



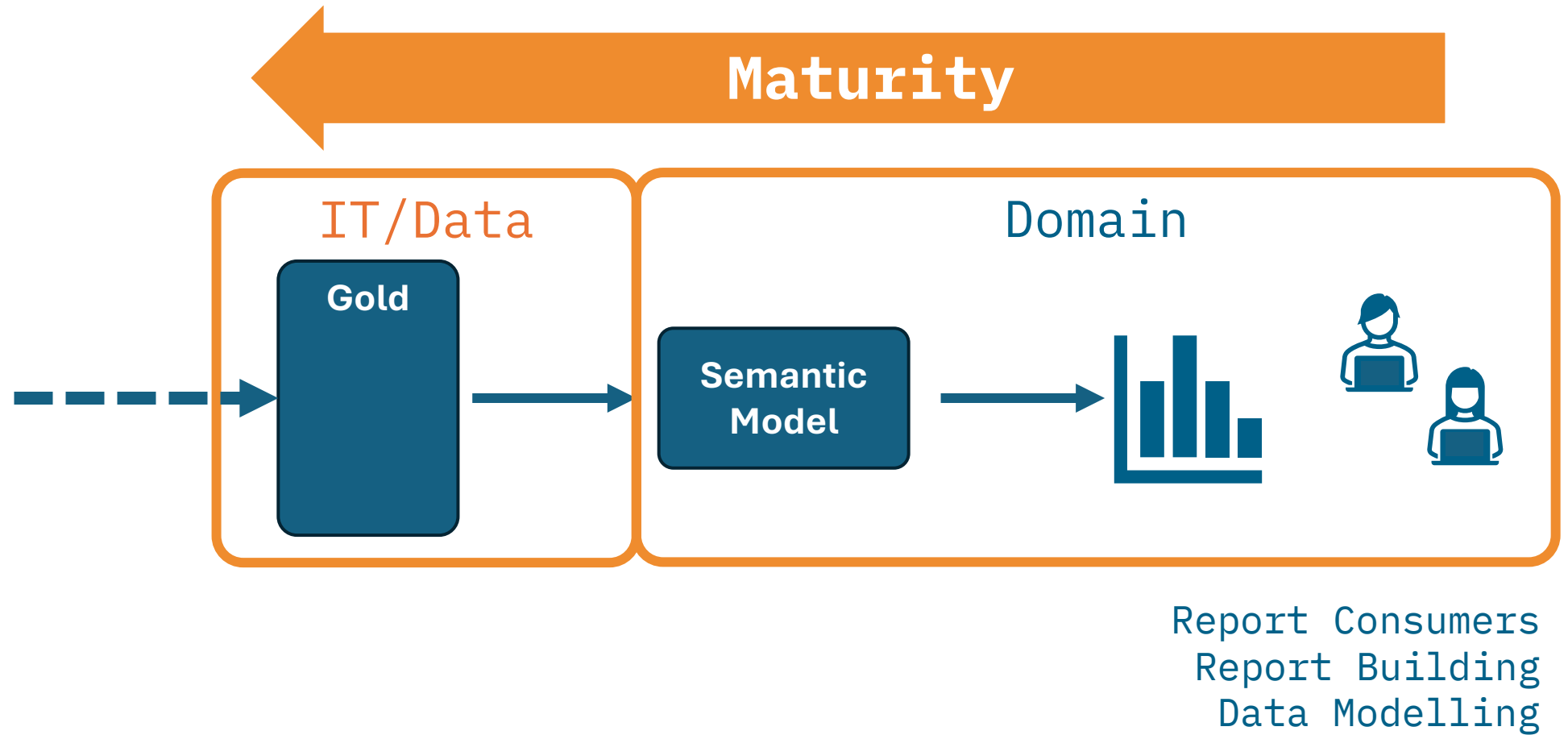
Assess Maturity

Data
Maturity



Assess Maturity

Data
Maturity



Assess Maturity

Data
Maturity

Maturity

Domain

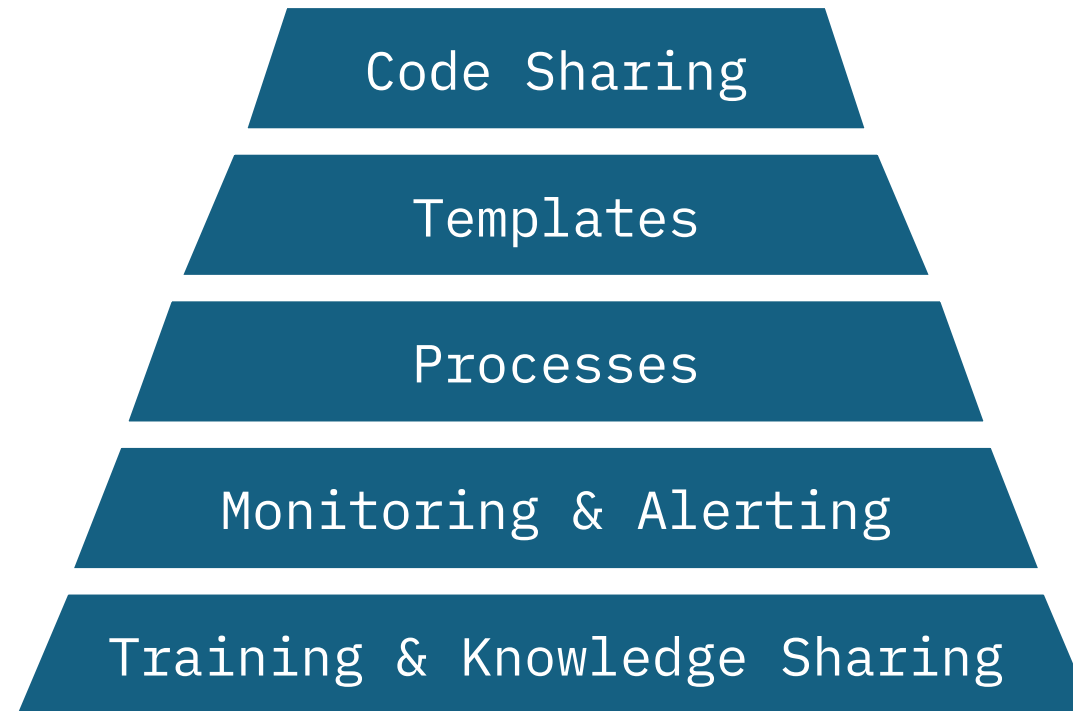
Gold

Semantic
Model



Report Consumers
Report Building
Data Modelling
Data Curation
Business Logic
Data Aggregations

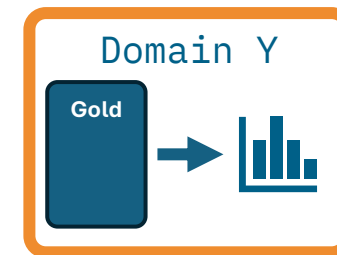
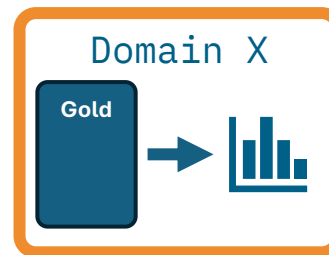
Provide Support



Automate

Technology

- **Infrastructure**
- **Security Roles**
- **Policies**
- **Tools**



4 Control > Guardrails



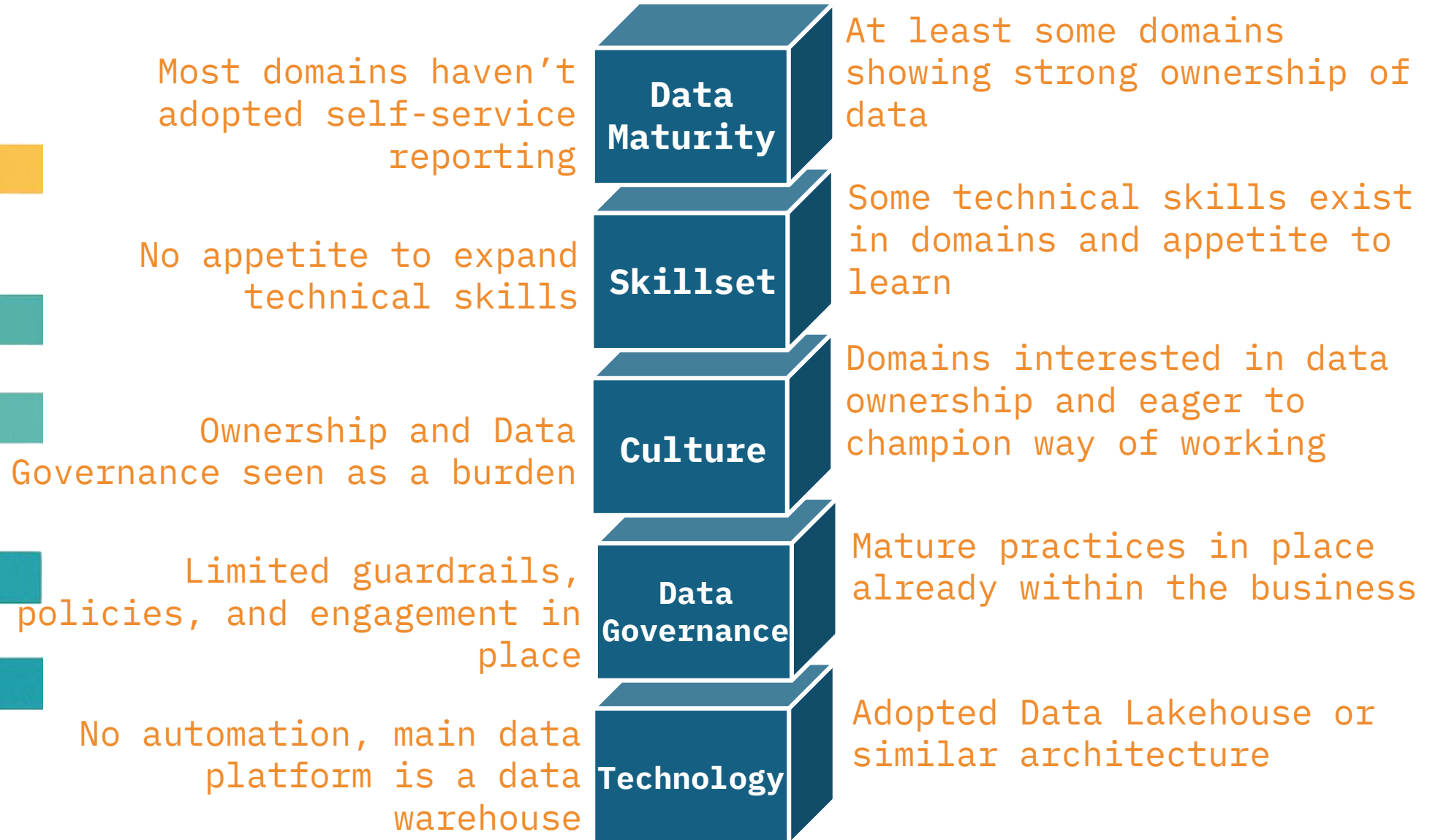
Control

Guardrail

Approval Gates	➔	Automated Policy Enforcement
Central Capacity & Costs	➔	Budgets & Throttling
Performance Protections	➔	Education and Training
Data Product ownership	➔	Certification and discoverability



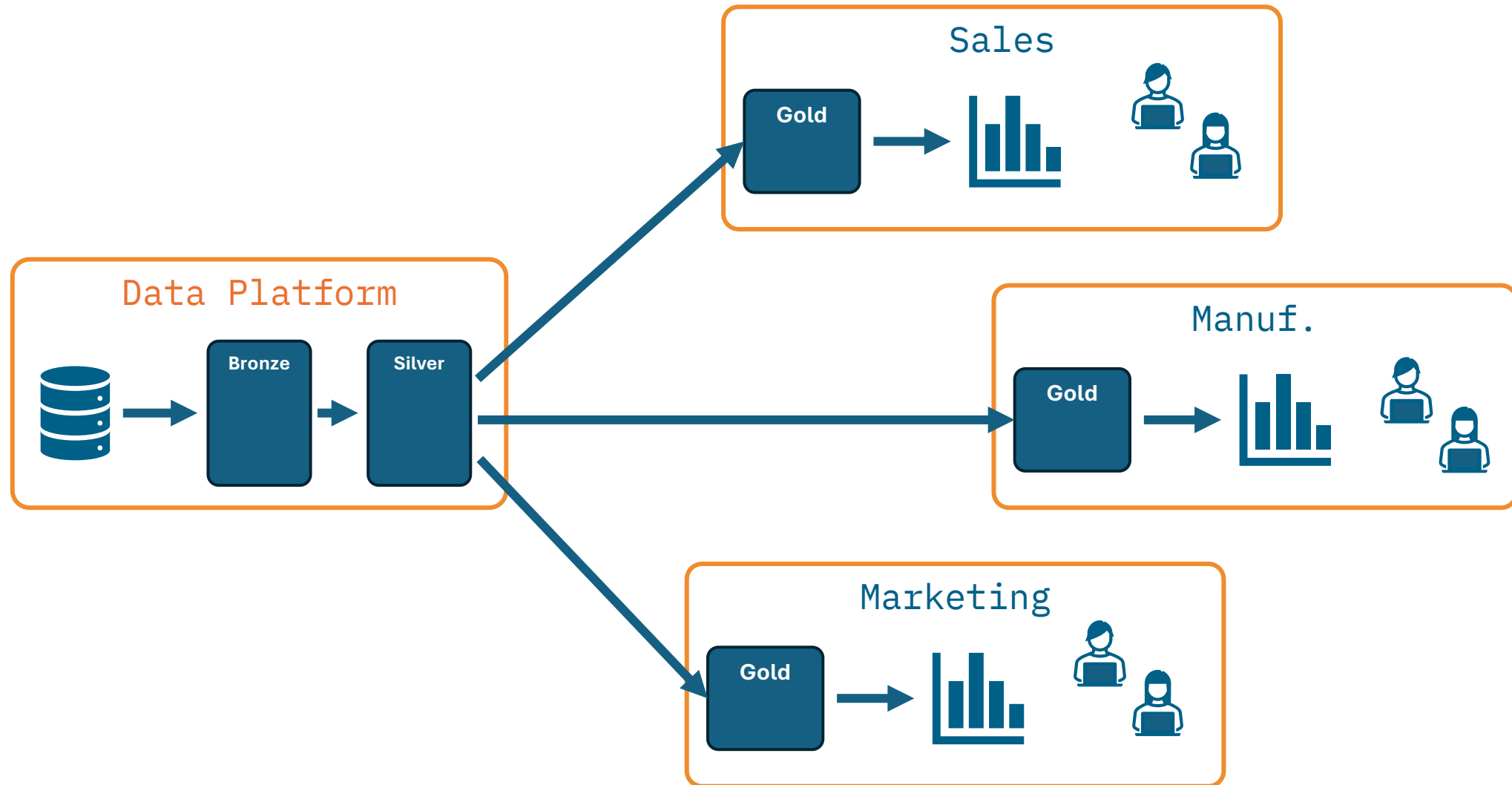
Centralise or Federate



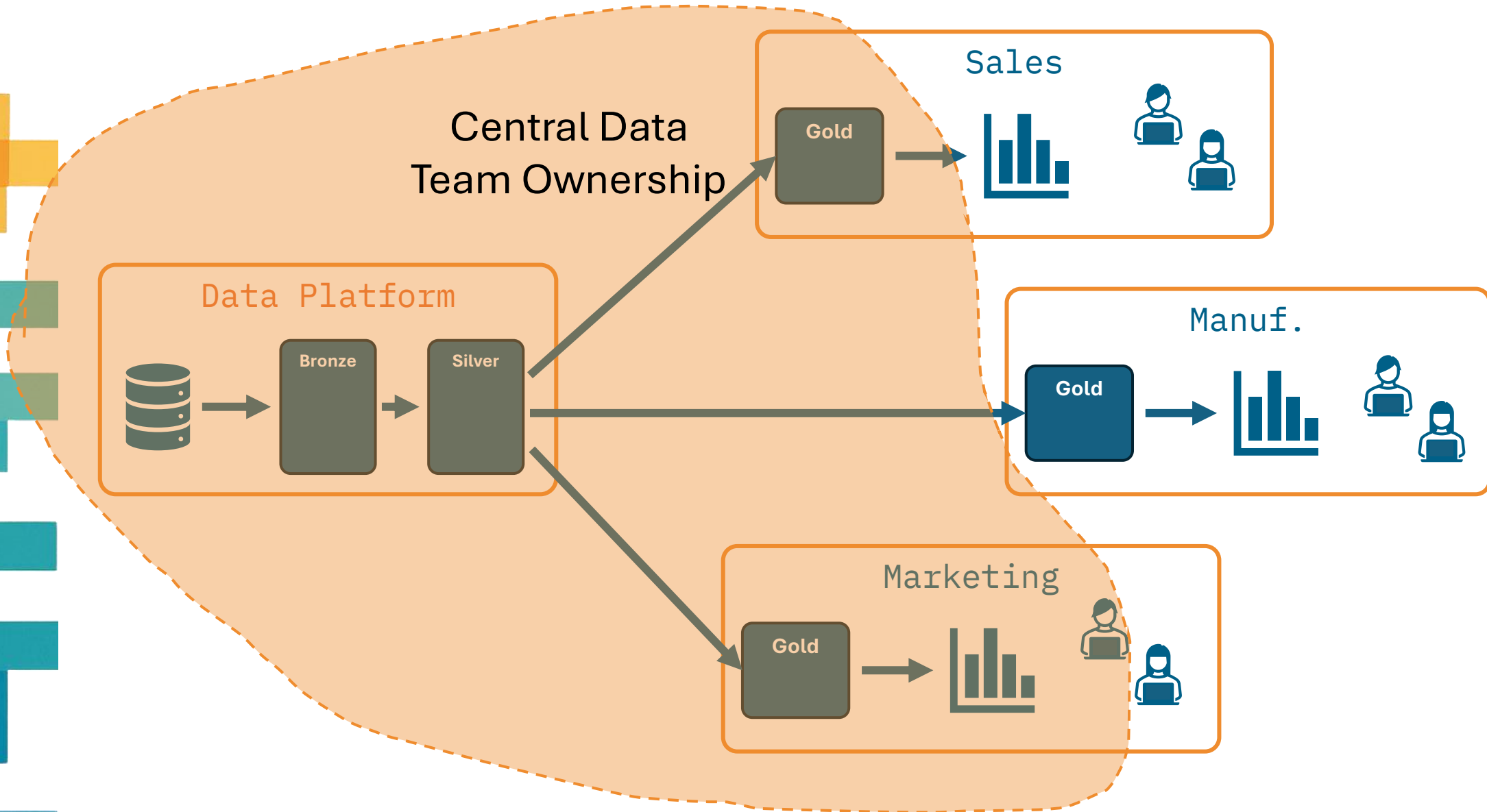


What does it look like?

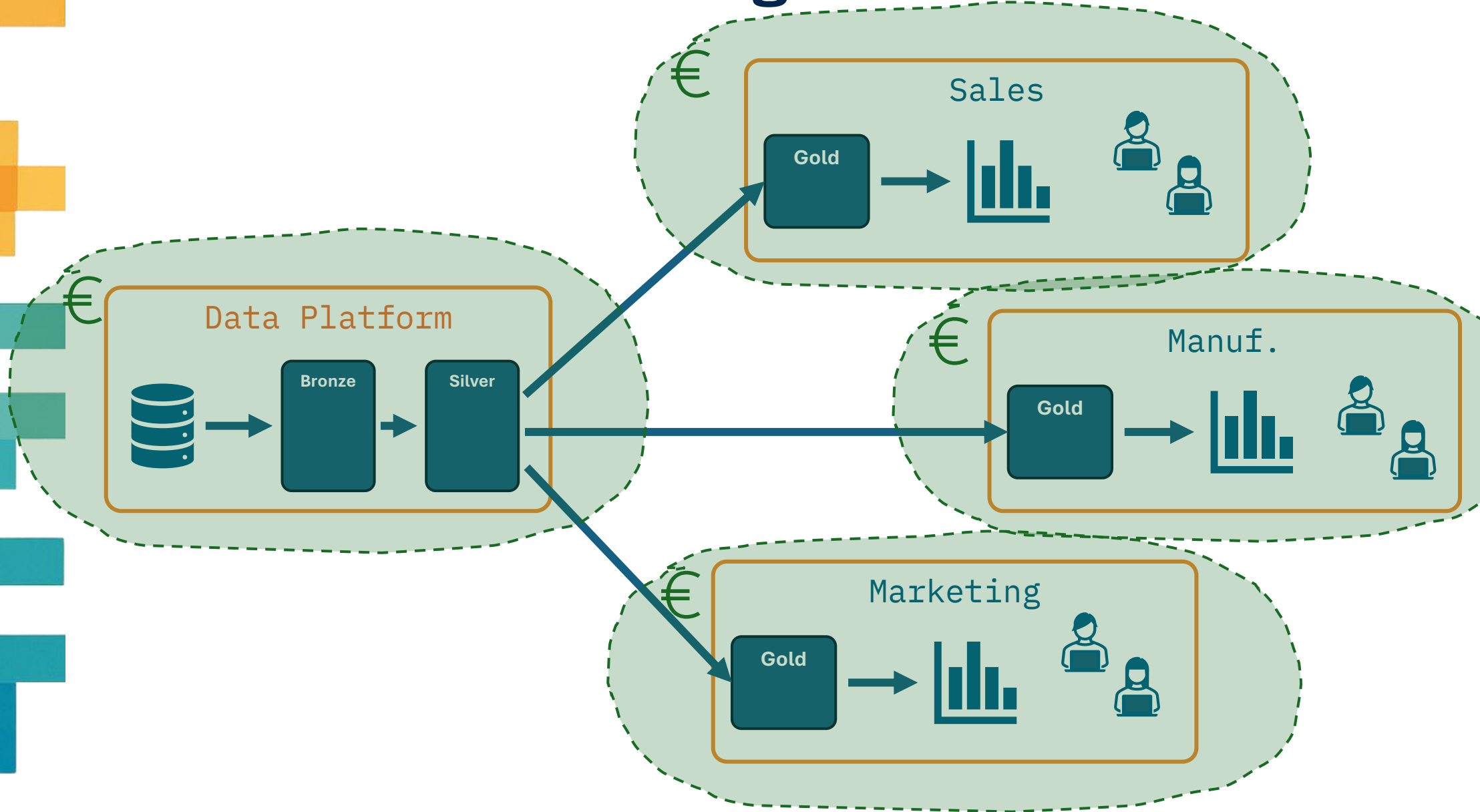
What we're working towards



What we're working towards



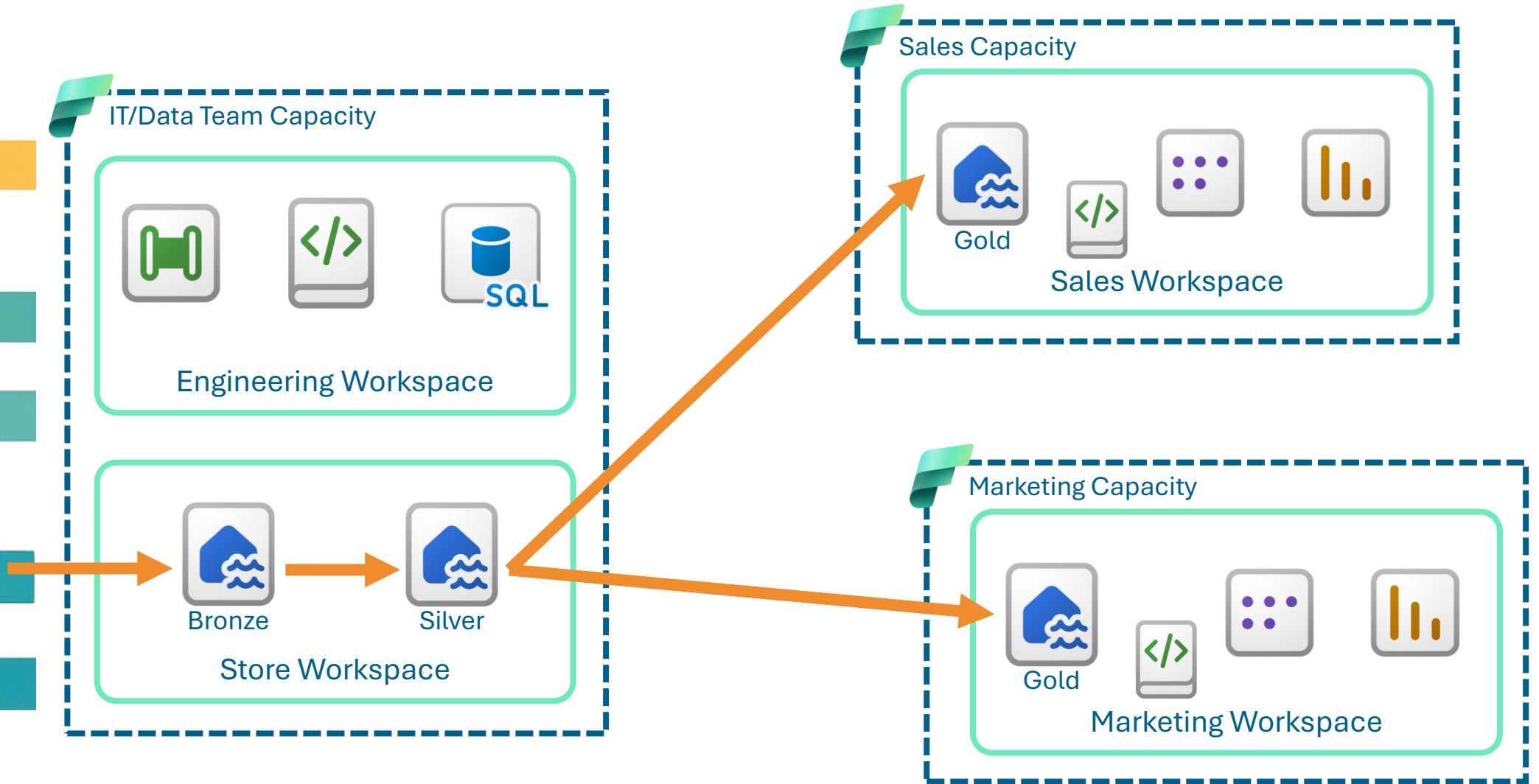
What we're working towards





The Fabric version

A Fabric Federated Data Mesh





Take aways

Anti-Patterns

- **“Fake federation”** → still central team doing everything
- **“Full decentralisation”** → no standards, data chaos
- **Tool-led architecture** → letting the platform dictate the model
- **Ignoring cost ownership** in federated setups



In Summary

**Assess your
Current
Model**

**Define
Ownership
Boundaries**

**Build
Guardrails,
not Gates**

Evolve Intentionally



Thank You



<https://craigporteous.com>



<https://github.com/cporteou>



<https://www.youtube.com/@cnporteous>

